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Heat exchanger for stirling machine, method for manufacturing heat exchanger, and stirling machine

Abstract

A heat exchanger for a Stirling machine that includes an expansion chamber and a compression chamber, the heat exchanger including: a heater; a regenerator; and a cooler, in which the heater includes a first flow channel communicating with the expansion chamber and the regenerator and configured to a working fluid to flow therethrough, and a second flow channel configured to allow a first fluid for exchanging heat with the working fluid to flow therethrough, the cooler includes a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and the heater, the regenerator, and the cooler are continuously connected to each other in series in a flow direction of the working fluid, provided in one piece.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application No. 2023-149576 filed on Sep. 14, 2023, the content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a heat exchanger for a Stirling machine, a method for manufacturing the heat exchanger, and a Stirling machine.

BACKGROUND ART

(3) In recent years, researches and developments have been conducted that contribute to an improvement in energy efficiency in order to allow more people to access affordable, reliable,

sustainable, and modern energy.

(4) A Stirling engine is known as a heat engine that performs conversion between heat energy and mechanical power. In the Stirling engine, a working gas is reciprocated between an expansion chamber on a heater side and a compression chamber on a cooler side via a heater, a regenerator, and a cooler to repeat a volume change of the expansion chamber and the compression chamber, thereby performing conversion between heat energy and mechanical power (for example, see JP2022-126185A).

(5) In JP2022-126185A, when manufacturing a heat exchanger including a heater, a regenerator, and a cooler, it is necessary to perform brazing, welding, or the like in order to couple these components, and it takes time and cost in manufacturing the heat exchanger. In addition, when manufacturing the regenerator, a process of stacking metal fibers is necessary, which particularly takes time.

(6) An object of the present invention is to provide a heat exchanger for a Stirling machine, a method for manufacturing the heat exchanger, and a Stirling machine, which can reduce the manufacturing time and the manufacturing cost.

SUMMARY OF INVENTION

(7) According to an aspect of the present invention, there is provided a heat exchanger for a Stirling machine that includes an expansion chamber and a compression chamber, the heat exchanger including: a heater; a regenerator; and a cooler, in which the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid, and communicate the expansion chamber with the compression chamber, the heater includes a first flow channel communicating with the expansion chamber and the regenerator and configured to flow the working fluid to flow therethrough, and a second flow channel configured to allow a first fluid for exchanging heat with the working fluid to flow therethrough, the cooler includes a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece.

(8) According to another aspect of the present invention, there is provided a method for manufacturing the heat exchanger, the method including: forming the heater, the cooler, and the regenerator in one piece by performing additive manufacturing using a metal powder.

(9) According to still another aspect of the present invention, there is provided a Stirling machine including: a heat exchanger including a heater, a regenerator, and a cooler, which are connected in series in a flow direction of a working fluid; and a cylinder including an expansion chamber connected to the heater and a compression chamber connected to the cooler, in which the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger, the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece, and the heat exchanger and the cylinder are continuously connected to each other, provided in one piece.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating a Stirling engine 1 as an example of a Stirling machine according to the invention.

(2) FIG. 2 is a cross-sectional view illustrating a configuration of main parts of a heat exchanger 6.

(3) FIG. 3 is a cross-sectional view taken along a line III-III in FIG. 2.

- (4) FIG. 4 is a diagram schematically illustrating a working gas flow channel **671** and a heat source fluid flow channel **672** provided in a heater **67**.
- (5) FIG. 5 is a diagram schematically illustrating a cross section of the working gas flow channel **671** and the heat source fluid flow channel **672** of the heater **67** that are defined and formed by a partition wall **60**.
- (6) FIG. 6A is a diagram illustrating a spiral flow generation portion **55** provided in the working gas flow channel **671** of the heater **67**.
- (7) FIG. 6B is a diagram illustrating the working gas flow channel **671** as viewed from a flow direction of a heat source fluid.
- (8) FIG. 7A is a diagram illustrating a flow (indicated by broken lines) of a heat source fluid flowing in a heat source fluid introduction chamber **80**.
- (9) FIG. 7B is a diagram illustrating a ceiling wall **27** as viewed from an expansion chamber **21** side.
- (10) FIG. 8 is a cross-sectional view taken along a line VIII-VIII in FIG. 2.
- (11) FIG. 9 is a diagram schematically illustrating a working gas flow channel **681** and a cooling fluid flow channel **682** provided in a cooler **68**.
- (12) FIG. 10 is a perspective view of a heat storage **71**.
- (13) FIG. 11 is a side view of the heat storage **71** as viewed in the direction of a white arrow in FIG. 10.
- (14) FIG. 12 is a perspective view of a unit structure **72** of the heat storage **71**.
- (15) FIG. 13 is a diagram illustrating a side element **73** of the heat storage **71** as viewed from a second direction (upper part) and as viewed from a first direction (lower part).
- (16) FIG. 14 is a top view of a regenerator **69** in a state where the heat storage **71** is accommodated in a housing **70**.
- (17) FIG. 15 is a side view of the regenerator **69** in a state where the heat storage **71** is accommodated in the housing **70**.
- (18) FIG. 16 is a side view of the heat exchanger **6**, the heat source fluid introduction chamber **80**, and a heat source fluid discharge channel **51** that are formed in one piece.

DESCRIPTION OF EMBODIMENTS

- (19) Hereinafter, a Stirling machine of an embodiment of the present invention will be described with reference to the accompanying drawings. Hereinafter, a β -type Stirling engine will be described as an example of the Stirling machine.
- (20) [Stirling Machine]
- (21) As illustrated in FIG. 1, a Stirling engine **1** includes a cylinder **2** in which a working gas is sealed, a displacer piston **3** and a power piston **4** inserted into the cylinder **2**, a piston crank mechanism **5** connected to the displacer piston **3** and the power piston **4**, and a heat exchanger **6** that raises and lowers a temperature of the working gas in the cylinder **2**. The working gas is, for example, high-pressure air, helium, or hydrogen.
- (22) The Stirling engine **1** is, for example, a hermetic engine, and an electric generator **11** that extracts a shaft rotation output of the piston crank mechanism **5** as electric power is disposed inside a crankcase.
- (23) The cylinder **2** has a substantially cylindrical shape in which one end thereof is provided with a ceiling wall **27** and the other end thereof is open. In the following description, an axial direction of the cylinder **2** is also referred to as an up-down direction, and the ceiling wall **27** side and the opening side in the axial direction are also referred to as an upper side and a lower side, respectively.
- (24) In the cylinder **2**, the displacer piston **3** is provided on the upper side, and the power piston **4** is provided on the lower side. An expansion chamber **21** in which the working gas has a high temperature is formed between the displacer piston **3** and the ceiling wall **27**, and a compression chamber **22** in which the working gas has a low temperature is formed between the displacer piston

3 and the power piston **4**. Hereinafter, the expansion chamber **21** and the compression chamber **22** in the cylinder **2** are also referred to as a working space **20**.

(25) The cylinder **2** includes a first cylinder **23** that is positioned on the upper side and a second cylinder **24** that is positioned on the lower side and has the same diameter as the first cylinder **23**. The first cylinder **23** and the second cylinder **24** are coaxially connected to each other by flange portions **25** and **26** thereof.

(26) A plurality of first communication portions **23a** communicating with a plurality of first inlet-outlets (inlet-outlets) **61a** of the heat exchanger **6** respectively, which will be described later, are provided on an outer peripheral surface of the first cylinder **23** on the upper side. The first communication portions **23a** are provided in the same number as the first inlet-outlets **61a**. Each first communication portion **23a** is formed in a slit shape having substantially the same shape as the corresponding first inlet-outlet **61a**, and is disposed in a manner of facing the corresponding first inlet-outlet **61a**. The plurality of first communication portions **23a** serve as inlet-outlets to the expansion chamber **21**.

(27) A plurality of second communication portions **23b** communicating with a plurality of second inlet-outlets (inlet-outlets) **61b** of the heat exchanger **6** respectively, which will be described later, are provided on the outer peripheral surface of the first cylinder **23** on the lower side. The second communication portions **23b** are provided in the same number as the second inlet-outlets **61b**. Each second communication portion **23b** is formed in a slit shape having substantially the same shape as the corresponding second inlet-outlet **61b**, and is disposed in a manner of facing the corresponding second inlet-outlet **61b**. The second communication portions **23b** serve as inlet-outlets of the compression chamber **22**.

(28) The displacer piston **3** and the power piston **4** are coaxially disposed in the cylinder **2**. The displacer piston **3** divides a space defined by the cylinder **2** and the power piston **4** into two in the up-down direction. In other words, the displacer piston **3** constitutes the expansion chamber **21** formed above the displacer piston **3** and the compression chamber **22** formed between the displacer piston **3** and the power piston **4**. An internal space of the expansion chamber **21** and the compression chamber **22** (that is, the working space **20**) is in communication with an internal space of the heat exchanger **6** via the plurality of first communication portions **23a** and the plurality of second communication portions **23b**, and has substantially the same pressure as an internal pressure of the heat exchanger **6**. Thus, the expansion chamber **21** and the compression chamber **22** communicate with each other via the heat exchanger **6**, and the working gas repeatedly reciprocates between the expansion chamber **21** and the compression chamber **22** through the heat exchanger **6** by the movement of the displacer piston **3** described later.

(29) The displacer piston **3** and the power piston **4** are connected to a crankshaft **10** via different connecting rods **30** and **40** respectively such that the power piston **4** has a phase of moving in the lead by 90 degrees (a phase difference of 90 degrees), and the power piston **4** outputs a rotational force to the piston crank mechanism **5** via the connecting rod **40**. The crankshaft **10** is coupled to an output shaft **16** of a recoil starter **15** via a coupling **14**.

(30) The displacer piston **3** is used to move the working gas in the cylinder **2**, and moves the working gas in the cylinder **2** to change a volume ratio between the expansion chamber **21** and the compression chamber **22**. For example, when a large amount of high-temperature working gas exists in the expansion chamber **21**, the internal pressure of the cylinder **2** is high, and when a large amount of low-temperature working gas exists in the compression chamber **22**, the internal pressure of the cylinder **2** is low. In accordance with timings of this pressure fluctuation, when the pressure is high, the power piston **4** is pushed down from near the top dead center to obtain a shaft rotational force to be output to the piston crank mechanism **5**, and when the pressure is low, the power piston **4** is pushed up by the piston crank mechanism **5**. By repeating such an operation, a continuous rotational force can be obtained.

(31) As described above, the Stirling engine **1** constitutes a Stirling cycle in which the working gas

is moved between the expansion chamber **21** and the compression chamber **22** via the heat exchanger **6** to repeat the volume change of the expansion chamber **21** and the compression chamber **22**, thereby performing conversion between heat and mechanical power.

(32) [Heat Exchanger]

(33) As illustrated in FIG. 2, the heat exchanger **6** is provided in a double cylindrical shape extending in the axial direction (that is, the up-down direction) along the outer peripheral surface of the first cylinder **23**, and has an inner peripheral wall **61** provided on an inner peripheral side and an outer peripheral wall **62** provided on an outer peripheral side.

(34) The heat exchanger **6** includes a heater **67** that is provided on the expansion chamber **21** side and performs heat exchange between the working gas and a heat source fluid, a cooler **68** that is provided on the compression chamber **22** side and performs heat exchange between the working gas and a cooling fluid, and a regenerator **69** that is disposed between the heater **67** and the cooler **68** and is capable of storing heat of the working gas. For example, the heat source fluid is a high-temperature gas (for example, high-temperature air), and the cooling fluid is a low-temperature liquid (for example, cooling water).

(35) The heater **67**, the regenerator **69**, and the cooler **68** are disposed between the inner peripheral wall **61** and the outer peripheral wall **62**, and are connected in series in the axial direction of the cylinder **2**.

(36) The heater **67**, the regenerator **69**, and the cooler **68** are provided with a communication path **600** through which the expansion chamber **21** and the compression chamber **22** communicate with each other and through which the working gas is circulated. The communication path **600** includes a plurality of working gas flow channels **671** of the heater **67**, an internal space of the regenerator **69**, and working gas flow channels **681** of the cooler **68**, which will be described later.

(37) [Heater]

(38) As illustrated in FIG. 3, the heater **67** includes a plurality of first partition walls **60a** that are provided at predetermined intervals in a circumferential direction and extend in a radial direction, and a plurality of second partition walls **60b** that connect facing surfaces of the first partition walls **60a** and extend in a flow direction of the working gas and the heat source fluid (that is, the up-down direction).

(39) In the heater **67**, working gas circulation layers **63** for circulating the working gas and heat source fluid circulation layers **64** for circulating the heat source fluid are alternately formed in the circumferential direction across the first partition wall **60a**. The heater **67** is a plate-type heat exchanger that performs heat exchange between the working gas and the heat source fluid via the first partition wall **60a**. In the present embodiment, since the temperature of the heat source fluid is higher than that of the working gas, the heater **67** heats the working gas going into and out of the expansion chamber **21** via the first communication portion **23a** of the first cylinder **23**.

(40) As illustrated in FIG. 2, the first inlet-outlet **61a** is provided at an upper end of the inner peripheral wall **61** where the working gas circulation layer **63** is positioned. The first inlet-outlet **61a** is formed in a slit shape having substantially the same shape as the corresponding first communication portion **23a**, and is disposed facing the corresponding first communication portion **23a** of the first cylinder **23**. In the heat exchanger **6**, the first inlet-outlet **61a** serves as an inlet-outlet to the expansion chamber **21** of the cylinder **2**.

(41) An upper end **631** of the working gas circulation layer **63** is closed, and a lower end thereof is open to be in communication with the regenerator **69**. The direction in which the working gas flowing through the working gas circulation layer **63** flows one time during one rotation of the crankshaft **10** due to the movement of the displacer piston **3** changes to the opposite direction.

(42) An upper end of the heat source fluid circulation layer **64** is open and communicates with a heat source fluid introduction chamber **80**. The heat source fluid supplied from the outside is introduced into the heat source fluid circulation layer **64** through the heat source fluid introduction chamber **80**. Accordingly, the heat source fluid is introduced into the heat source fluid circulation

layer **64** from above. Details of the heat source fluid introduction chamber **80** will be described later.

(43) An outlet **62a** is provided in the outer peripheral wall **62** positioned at a lower end of the heat source fluid circulation layer **64**. The outlet **62a** is formed in a slit shape, is disposed facing the corresponding heat source fluid circulation layer **64**, and guides the heat source fluid to a heat source fluid discharge channel **51** provided on the outer side of the heat source fluid circulation layer **64** in the radial direction. An abutting portion **641** is provided at the lower end of the heat source fluid circulation layer **64**. The abutting portion **641** is provided along a lower end of the outlet **62a** of the outer peripheral wall **62**, and guides the heat source fluid to the outlet **62a**.

(44) The heat source fluid discharge channel **51** is provided substantially at the center of the heat exchanger **6** in the up-down direction, and extends along the circumferential direction of the heat exchanger **6**. The heat source fluid discharge channel **51** is provided with a discharge portion **52**, and the heat source fluid guided from each outlet **62a** to the heat source fluid discharge channel **51** is discharged from the discharge portion **52** to the outside of the heat exchanger **6**.

(45) To describe an inner structure of the working gas circulation layer **63** and the heat source fluid circulation layer **64** in more detail, the heater **67** includes a plurality of working gas flow channels **671** that are provided in the working gas circulation layer **63** and through which the working gas is circulated, and a plurality of heat source fluid flow channels **672** that are provided in the heat source fluid circulation layer **64** and through which the heat source fluid is circulated. Each working gas flow channel **671** and each heat source fluid flow channel **672** are defined and formed by the first partition wall **60a** and the second partition wall **60b**. The first partition wall **60a** and the second partition wall **60b** that define and form each flow channel may be collectively referred to as a partition wall **60**.

(46) FIG. **4** is a diagram schematically illustrating the working gas flow channel **671** and the heat source fluid flow channel **672**, and illustration of the partition wall **60** is omitted in FIG. **4**. The working gas flow channel **671** extends in the up-down direction, and an upper end thereof communicates with the expansion chamber **21** via a space **630** and the first inlet-outlet **61a** described later, and a lower end thereof communicates with the regenerator **69**. In each working gas circulation layer **63**, a plurality of working gas flow channels **671** are disposed in parallel along the radial direction. The working gas flows in a counter flow to the heat source fluid when flowing from the lower end to the upper end of the working gas flow channel **671**, and the working gas flows in a parallel flow with the heat source fluid when flowing from the upper end to the lower end.

(47) The upper end of the working gas flow channel **671** is positioned farther from the closed upper end **631** of the working gas circulation layer **63** as the working gas flow channel **671** is positioned closer to the first inlet-outlet **61a** (that is, on the inner peripheral side). That is, the working gas discharged from the working gas flow channel **671** positioned closer to the first inlet-outlet **61a** is discharged at a position farther from the upper end **631** of the working gas circulation layer **63**.

(48) In each working gas circulation layer **63** of the heater **67**, the space **630** defined by the upper end **631** and the upper ends of the plurality of working gas flow channels **671** has a substantially triangular shape in a side view. The space **630** is a portion where the flow of the working gas flowing out of the working gas flow channel **671** is bent toward the first inlet-outlet **61a**. The space **630** has a shape in which a flow channel cross section thereof gradually increases from the outer peripheral side toward the inner peripheral side, and therefore, flow velocities at respective positions are equal and a pressure loss can be minimized. By setting a maximum area of the first inlet-outlet **61a** to be equal to the sum of cross-sectional areas of the plurality of working gas flow channels **671**, the pressure loss of the heat exchanger **6** including the first inlet-outlet **61a** can be minimized.

(49) FIG. **5** is a diagram schematically illustrating a plurality of working gas flow channels **671** provided in each working gas circulation layer **63**. As illustrated in FIG. **5**, the plurality of working

gas flow channels **671** are defined and formed by the first partition wall **60a** and the second partition wall **60b**. At this time, the plurality of working gas flow channels **671** are partitioned by the second partition wall **60b** so as to have the same equivalent diameter. By making the equivalent diameters of the respective working gas flow channels **671** equal, a pressure loss coefficient and a heat transfer coefficient become equal, and a drift of the working gas flowing through the respective working gas flow channels **671** is eliminated. By eliminating the drift of the working gas flowing through the working gas flow channel **671**, the overall pressure loss is reduced, and the heat transfer performance is improved.

(50) Returning to FIG. 2, the heat source fluid flow channel **672** extends in the up-down direction of the heat exchanger **6**, and has an upper end communicating with the heat source fluid introduction chamber **80** and a lower end communicating with the heat source fluid discharge channel **51** via a space **640** and the outlet **62a** described later. In each heat source fluid circulation layer **64**, a plurality of heat source fluid flow channels **672** are disposed in parallel along the radial direction.

(51) The lower end of the heat source fluid flow channel **672** is positioned farther from the abutting portion **641** as the heat source fluid flow channel **672** is positioned closer to the outlet **62a** (that is, on the outer peripheral side). That is, the heat source fluid discharged from the heat source fluid flow channel **672** positioned closer to the outlet **62a** is discharged at a position farther from the abutting portion **641**.

(52) In each heat source fluid circulation layer **64**, the space **640** defined by the abutting portion **641** and the lower ends of the plurality of heat source fluid flow channels **672** has a substantially triangular shape in a side view. The space **640** is a portion where the flow of the heat source fluid flowing out of the heat source fluid flow channel **672** is bent toward the outlet **62a**. The space **640** has a shape in which a flow channel cross section thereof gradually increases from the inner peripheral side toward the outer peripheral side, and therefore, flow velocities at respective positions are equal and a pressure loss can be minimized.

(53) As indicated by the reference signs in parentheses in FIG. 5, a plurality of heat source fluid flow channels **672** provided in each heat source fluid circulation layer **64** are also defined and formed by the first partition wall **60a** and the second partition wall **60b** similarly to the working gas flow channels **671**.

(54) In the present embodiment, the heat conductivity of the heat source fluid is lower than the heat conductivity of the working gas, and the heat transfer coefficient in a flow in which the heat source fluid is circulated through the heat source fluid flow channel **672** is smaller than the heat transfer coefficient in a flow in which the working gas is circulated through the working gas flow channel **671**. In other words, the heat resistance according to the heat transfer of the heat source fluid (hereinafter, also simply referred to as the heat resistance on the heat source fluid side) is larger.

(55) As illustrated in FIGS. 6A and 6B, the heat source fluid flow channel **672** is provided with a spiral flow generation portion **55** that generates a spiral flow (indicated by a solid arrow in FIG. 6A) along the flow direction of the heat source fluid (that is, a downward direction). Accordingly, a tornado-like vortex is generated along the flow direction of the heat source fluid, and the heat source fluid flowing through the heat source fluid flow channel **672** is stirred. Therefore, the heat resistance on the heat source fluid side is reduced, and the heat transfer performance of the working gas and the heat source fluid via the partition wall **60** is improved.

(56) Specifically, the spiral flow generation portion **55** includes a plurality of fins **56** erected from the partition wall **60**. The plurality of fins **56** are disposed at predetermined intervals along the flow direction of the heat source fluid, and are spirally provided around a central axis **L1** of the flow channel cross section of the heat source fluid flow channel **672**. FIGS. 6A and 6B illustrate an example in which the flow channel cross section of the heat source fluid flow channel **672** has a rectangular shape, and the plurality of fins **56** are spirally provided at intervals of 90 degrees in the circumferential direction along the central axis **L1**. The flow channel cross section of the heat

source fluid flow channel **672** may be any shape. The angular interval in the circumferential direction of the plurality of fins **56** is not limited to 90 degrees. It is sufficient that the fins are provided spirally at angular intervals of 30 degrees or greater and 180 degrees or smaller in the circumferential direction.

(57) Each fin **56** has a base end **56a** connected to the partition wall **60**. The base end **56a** is connected to the partition wall **60** in a manner of inclining at a predetermined angle with respect to a plane perpendicular to the central axis **L1** so as to generate a spiral flow of the heat source fluid. As compared with a case where the fins **56** are not provided, the pressure loss of the heat source fluid flowing through the heat source fluid flow channel **672** is increased due to provision of the fins **56**, but the degree of increase in the pressure loss can be reduced by connecting the fins **56** to the partition wall **60** in a manner of inclining at a predetermined angle. Even when a foreign matter enters the heat source fluid flow channel **672** together with the heat source fluid, the foreign matter is less likely to accumulate on the fins **56**, and thus it is possible to prevent clogging of the heat source fluid flow channel **672**.

(58) The fins **56** adjacent to each other in the flow direction of the heat source fluid are provided at positions not overlapping each other in the flow direction. In other words, a downstream end **56b** of the fin **56** positioned upstream is positioned upstream of an upstream end **56c** of the fin **56** positioned downstream. Accordingly, the degree of increase in the pressure loss when the heat source fluid flows through the heat source fluid flow channel **672** can be further reduced. Further, it is possible to further prevent clogging of the heat source fluid flow channel **672** involving entry of a foreign matter.

(59) As illustrated in FIG. **6B**, the heat source fluid flow channel **672** includes a pass-through region **57**, in which the fins **56** are not provided, as viewed from the flow direction of the heat source fluid. In the present embodiment, as viewed from the flow direction of the heat source fluid, a region including the central axis **L1** of the heat source fluid flow channel **672** is the pass-through region **57**. Accordingly, the degree of increase in the pressure loss when the heat source fluid flows through the heat source fluid flow channel **672** can be further reduced.

(60) As described above, when the spiral flow generation portion **55** is provided in the flow channel, the heat transfer coefficient increases and the heat resistance decreases. On the other hand, as compared with a case where the spiral flow generation portion **55** is not provided, the pressure loss of the fluid increases, and the output performance of the Stirling engine **1** may be affected. Accordingly, it is preferable to determine whether to provide the spiral flow generation portion **55** in the flow channel in consideration of the balance between the effect of the decrease in the heat resistance and the influence of the increase in the pressure loss due to providing the spiral flow generation portion **55** in the flow channel.

(61) Among the working gas, the heat source fluid, and the cooling fluid flowing through the heat exchanger **6**, the working gas and the cooling fluid have high heat conductivity. Further, the heat transfer coefficient in a flow in which the working gas is circulated through the working gas flow channel **671** of the heater **67** and the working gas flow channel **681** of the cooler **68** described later, and the heat transfer coefficient in a flow in which the cooling fluid is circulated through the cooling fluid flow channel **682** of the cooler **68** described later are all large. Thus, the heat resistance on the working gas side and the cooling fluid side is small. At this time, the effect of reducing the heat resistance by providing the spiral flow generation portion **55** in the working gas flow channel **671** of the heater **67** and the working gas flow channel **681** and the cooling fluid flow channel **682** of the cooler **68** is relatively insignificant. Accordingly, in consideration of the influence of the increase in the pressure loss, it is preferable that the spiral flow generation portion **55** is not provided in the working gas flow channel **671** of the heater **67** and the working gas flow channel **681** and the cooling fluid flow channel **682** of the cooler **68**.

(62) On the other hand, as described above, the heat resistance on the heat source fluid side is large, and the effect of reducing the heat resistance by providing the spiral flow generation portion **55** in

the heat source fluid flow channel **672** is significant. Accordingly, even in consideration of the influence of the increase in the pressure loss, it is preferable to improve the heat transfer performance by providing the spiral flow generation portion **55** in the heat source fluid flow channel **672**.

(63) Next, the heat source fluid introduction chamber **80** will be described with reference to FIGS. 7A and 7B. The heat source fluid introduction chamber **80** is provided on the outer side (the upper side in the present embodiment) of the expansion chamber **21**, and the heat source fluid introduction chamber **80** and the expansion chamber **21** are partitioned by the ceiling wall **27** of the cylinder **2**.

(64) The heat source fluid introduction chamber **80** includes an introduction port **81** through which the heat source fluid is introduced from the outside into the heat source fluid introduction chamber **80**, and a guide port **82** serving as an inlet of the heat source fluid from the heat source fluid introduction chamber **80** to the heat source fluid flow channel **672**. The introduction port **81** is provided in an upper end of the heat source fluid introduction chamber **80**, and the guide port **82** is provided in communication with the upper end of the heat source fluid flow channel **672**.

(65) Since the heat source fluid introduction chamber **80** and the expansion chamber **21** are partitioned by the ceiling wall **27**, the heat exchange between the heat source fluid flowing through the heat source fluid introduction chamber **80** and the working gas flowing through the expansion chamber **21** is performed via the ceiling wall **27**. Accordingly, the working gas on the expansion chamber **21** side is subjected to heat exchange with the heat source fluid not only when the working gas flows through the working gas flow channel **671** but also when the working gas is present in the expansion chamber **21**, and therefore, a large amount of heat energy can be taken in from the heat source fluid.

(66) The ceiling wall **27** is provided with a plurality of grooves **29** on the expansion chamber **21** side. The plurality of grooves **29** are not a result of surface roughness formed at the time of manufacturing but formed by design. Although FIG. 7B illustrates an example in which the plurality of grooves **29** are provided on the expansion chamber **21** side, the plurality of grooves **29** may be provided on the outer side of the expansion chamber **21**, that is, on the heat source fluid introduction chamber **80** side, or may be provided on both the expansion chamber **21** side and the heat source fluid introduction chamber **80** side.

(67) The plurality of grooves **29** are provided in the entire region of the ceiling wall **27** and extend from a center side of the ceiling wall **27** toward an outer peripheral side (the heater **67** side).

(68) Since a surface area of the ceiling wall **27** is increased by providing the plurality of grooves **29** in the ceiling wall **27**, the working gas can take more heat energy from the heat source fluid, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall **27** is improved. As a result, the heat efficiency and output performance of the Stirling engine **1** can be improved. The plurality of grooves **29** are not necessarily provided in the entire region of the ceiling wall **27**, and may be provided in parts of the ceiling wall **27**.

(69) FIG. 7B illustrates an example in which the plurality of grooves **29** linearly extend from the center side to the outer peripheral side, and the present invention is not limited thereto. The plurality of grooves **29** may extend from the center side to the outer peripheral side in, for example, a zigzag shape (a shape in which the extending direction thereof changes in the middle), or may extend from the center side to the outer peripheral side in a curved shape. Further, depending on the position of the ceiling wall **27**, the plurality of grooves **29** may be a combination of linear, zigzag, and curved grooves. Thus, more heat energy can be taken in by appropriately changing the shape of the plurality of grooves **29**.

(70) The ceiling wall **27** is provided with a substantially conical protruding portion **28** protruding toward the expansion chamber **21**. The protruding portion **28** is formed by recessing a portion of the ceiling wall **27** toward the expansion chamber **21**, and is provided such that a central axis thereof coincides with the central axis of the cylinder **2**. Since the protruding portion **28** is provided

on the ceiling wall **27**, the surface area of the ceiling wall **27** can be increased as compared with a case where the ceiling wall **27** is formed in a flat plate shape. Accordingly, the heat energy taken in by the working gas can be further increased, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall **27** is further improved. Further, by providing the protruding portion **28**, the pressure resistance of the ceiling wall **27** is also increased.

(71) The heat source fluid introduction chamber **80** is provided with a guide portion **83** that guides the heat source fluid toward an apex part **28a** of the protruding portion **28** of the ceiling wall **27**. The guide portion **83** extends along the central axis of the protruding portion **28** to a position overlapping the protruding portion **28** in the axial direction of the cylinder **2**. As indicated by broken lines in FIG. 7A, the guide portion **83** guides the heat source fluid flowing in from the introduction port **81** to the apex part **28a** of the ceiling wall **27**. The heat source fluid guided to the apex part **28a** by the guide portion **83** flows along the substantially conical protruding portion **28** to the outer side in the radial direction, and is introduced into the heat source fluid flow channel **672** from the guide port **82**. Accordingly, the heat source fluid easily flows along the ceiling wall **27**, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall **27** can be further improved.

(72) Since the plurality of grooves **29** extend from the apex part **28a** of the protruding portion **28** toward the heater **67** side, the heat source fluid can smoothly flow from the apex part **28a** toward the heater **67** side.

(73) [Cooler]

(74) As illustrated in FIG. 8, similarly to the heater **67**, the cooler **68** includes a plurality of first partition walls **60a** that are provided at predetermined intervals in the circumferential direction and extend in the radial direction.

(75) As illustrated in FIGS. 2 and 8, in the cooler **68**, working gas circulation layers **65** for circulating the working gas and cooling fluid circulation layers **66** for circulating the cooling fluid are alternately formed in the circumferential direction across the first partition wall **60a**. The cooler **68** is a plate-type heat exchanger that performs heat exchange between the working gas and the cooling fluid via the first partition wall **60a**. In the present embodiment, since the temperature of the cooling fluid is lower than that of the working gas, the cooler **68** cools the working gas flowing into and out of the compression chamber **22** via the second communication portion **23b** of the first cylinder **23**.

(76) The second inlet-outlet **61b** described above is provided at the lower end of the inner peripheral wall **61** where the working gas circulation layer **65** is positioned. The second inlet-outlet **61b** is formed in a slit shape having substantially the same shape as the corresponding second communication portion **23b**, and is disposed facing the corresponding second communication portion **23b** of the first cylinder **23**. In the heat exchanger **6**, the second inlet-outlet **61b** serves as an inlet-outlet to the compression chamber **22** of the cylinder **2**.

(77) A lower end **651** of the working gas circulation layer **65** is closed, and an upper end thereof is open to be in communication with the regenerator **69**. The direction in which the working gas flowing through the working gas circulation layer **65** flows one time during one rotation of the crankshaft **10** due to the movement of the displacer piston **3** changes to the opposite direction.

(78) To describe an inner structure of the working gas circulation layer **65** and the cooling fluid circulation layer **66** in more detail, the cooler **68** includes a plurality of working gas flow channels **681** that are provided in the working gas circulation layer **65** and through which the working gas is circulated, and a plurality of cooling fluid flow channels **682** that are provided in the cooling fluid circulation layer **66** and through which the cooling fluid is circulated. In the present embodiment, two cooling fluid flow channels **682** are provided side by side in the circumferential direction in each cooling fluid circulation layer **66**, and are partitioned by a third partition wall **60c** extending in the radial direction. Similarly to the working gas flow channels **671** and the heat source fluid flow channels **672**, the working gas flow channels **681** are also defined and formed by the first partition

wall **60a** and the second partition wall **60b**. Since the configuration of the working gas flow channel **681** is similar to the configuration of the working gas flow channel **671** provided in the heater **67**, a detailed description thereof will be omitted.

(79) FIG. **9** is a diagram schematically illustrating the working gas flow channel **681** and the cooling fluid flow channel **682** provided in the cooler **68**. In FIG. **9**, the partition wall **60** is not shown.

(80) The working gas flow channel **681** extends in the up-down direction, and a lower end thereof communicates with the compression chamber **22** via a space **650** and the second inlet-outlet **61b** described later, and an upper end thereof communicates with the regenerator **69**. In each working gas circulation layer **65**, a plurality of working gas flow channels **681** are disposed in parallel along the radial direction.

(81) The lower end of the working gas flow channel **681** is positioned farther from the closed lower end **651** of the working gas circulation layer **65** as the working gas flow channel **681** is positioned closer to the second inlet-outlet **61b** (that is, on the inner peripheral side). That is, the working gas discharged from the working gas flow channel **681** positioned closer to the second inlet-outlet **61b** is discharged at a position farther from the lower end **651** of the working gas circulation layer **65**.

(82) Also in the working gas circulation layer **65** of the cooler **68**, the space **650** defined by the lower end **651** and the lower ends of the plurality of working gas flow channels **681** has a substantially triangular shape in a side view. Similarly to the plurality of working gas flow channels **671** in each working gas circulation layer **63**, the plurality of working gas flow channels **681** in each working gas circulation layer **65** are formed by partitioning a space between a pair of first partition walls **60a** by the second partition walls **60b**. Therefore, the description of the working gas flow channel **671** is referenced to, and is omitted here.

(83) Each cooling fluid flow channel **682** is formed by folding back a tubular member extending in the radial direction at one end (for example, an end on the outer peripheral side) in the radial direction and then folding back the tubular member at the other end (for example, an end on the inner peripheral side), which is repeated. Each cooling fluid flow channel **682** is formed in a so-called zigzag shape. Each cooling fluid flow channel **682** is formed by bending one flow channel into a folded state instead of being formed with a plurality of linear flow channels, and thus the contact surface of the cooling fluid with the cooling fluid flow channel **682** can be increased.

(84) As illustrated in FIG. **1**, the cooling fluid flow channel **682** includes an inlet **683** provided on the lower end side in the up-down direction and an outlet **684** provided on the upper end side, and is configured such that the cooling fluid flows upward as a whole.

(85) When the working gas heated by the heater **67** flows from the upper end side to the lower end side, that is, into the cooler **68**, the working gas flowing through the working gas flow channel **681** has a temperature distribution in which the temperature decreases from the upper end side to the lower end side, that is, from the expansion chamber **21** side to the compression chamber **22** side, and the cooling fluid flowing through the cooling fluid flow channel **682** has a temperature distribution in which the temperature increases from the lower end side to the upper end side. In other words, when the working gas flows from the upper end side to the lower end side, the working gas and the cooling fluid flow in the working gas flow channel **681** and the cooling fluid flow channel **682** so as to have a counter flow temperature distribution as a whole, and thus high heat exchange efficiency can be achieved in the cooler **68**. Further, since the contact area of the working gas and the cooling fluid is increased by the cooling fluid flow channel **682** formed in the zigzag shape, the heat exchange efficiency is further improved in the cooler **68**. When the working gas flows from the lower end side to the upper end side, the working gas and the cooling fluid flow in the working gas flow channel **681** and the cooling fluid flow channel **682** so as to have a parallel flow temperature distribution as a whole.

(86) [Regenerator]

(87) When the working gas moves from the high-temperature expansion chamber **21** to the low-

temperature compression chamber **22**, the regenerator **69** cools the working gas by taking in the heat energy of the working gas (that is, storing heat). Conversely, when the working gas moves from the low-temperature compression chamber **22** to the high-temperature expansion chamber **21**, the regenerator **69** heats the working gas by imparting the stored heat energy to the working gas (that is, releasing heat).

(88) As illustrated in FIG. **2**, the regenerator **69** includes a heat storage **71** capable of storing heat of the working gas and a housing **70** surrounding the heat storage **71**.

(89) The housing **70** includes the inner peripheral wall **61** and the outer peripheral wall **62** described above, and is provided in a double cylindrical shape. The heat storage **71** is accommodated in an internal space of the housing **70**.

(90) In the present embodiment, the heat storage **71** and the housing **70** are continuously connected to each other and provided in one piece. Specifically, the regenerator **69** is an additive manufactured body in which metal powders are accumulated, and the heat storage **71** and the housing **70** are provided in one piece by metal additive manufacturing. According to such a configuration, it is possible to adopt a complicated structure having high heat exchange performance as compared with a regenerator in the related art that is formed by integrating a metal mesh, metal foam, or the like as a heat storage into a housing. Further, the number of assembly steps can be reduced.

(91) Hereinafter, a structure of the heat storage **71** will be described with reference to FIGS. **10** to **15**. The structure of the heat storage **71** illustrated in FIGS. **10** to **15** is an example, and the structure of the heat storage **71** can take various shapes. In the following description, an up-down direction in the drawings, that is, a flow direction of the working gas is also referred to as a first direction, and a horizontal direction in the drawings, that is, a direction intersecting the first direction is also referred to as a second direction.

(92) The heat storage **71** includes a plurality of side elements **73**, and the plurality of side elements **73** are continuously connected to each other in a three-dimensional manner and provided in one piece.

(93) More specifically, as illustrated in FIGS. **10** and **11**, the heat storage **71** includes a plurality of unit structures **72** each formed by connecting a plurality of side elements **73** to each other, and the unit structures **72** are continuously provided in the flow direction of the working gas and the direction intersecting therewith. The example illustrated in FIGS. **10** and **11** illustrates a configuration in which the unit structure **72** has a three-dimensional shape formed by connecting, by the side elements **73**, atoms disposed at positions excluding vertices in a unit cell of a diamond structure. The unit structure **72** may have various structures, for example, a structure formed by connecting, by the side elements **73**, vertices of a polyhedron such as a regular dodecahedron or a regular icosahedron.

(94) Since the plurality of side elements **73** of the heat storage **71** are continuously connected to each other in a three-dimensional manner and provided in one piece, it is possible to reduce a dead volume which is a volume not involved in heat exchange in the housing **70**, and to improve the heat storage performance of the regenerator **69**.

(95) Each side element **73** has a plate shape. Each side element **73** has a thickness of, for example, 0.1 mm or less, and is formed finely. Since it is necessary to repeat heat storage and heat release in a short time when exchanging heat energy with the working gas, each side element **73** is finely formed in this manner to reduce its heat capacity.

(96) As illustrated in FIGS. **12** and **13**, in each side element **73**, a surface **731** having a maximum surface area is provided along the first direction. That is, each side element **73** is provided such that a projected area **S2** as viewed from the second direction is larger than a projected area **S1** as viewed from the first direction.

(97) Since the projected area **S1** in the first direction is small, the flow of the working gas is prevented from being hindered by the side elements **73**, and the pressure loss at the time when the

working gas flows in the first direction can be reduced. Further, since the projected area S2 in the second direction is large, the contact surface along the flow of the working gas is large, and the heat storage amount of the heat storage 71 can be increased. Accordingly, with the side elements 73, it is possible to achieve both a reduction in pressure loss and an improvement in heat exchange efficiency.

(98) In each side element 73, in a cross section cut along the first direction, an end 732 on the heater 67 side and an end 733 on the cooler 68 side are provided in a curved shape. Specifically, in each side element 73, a cross section 73s cut along the first direction has an elliptical shape as illustrated in FIG. 12, and both ends of the major axis of the ellipse correspond to the ends 732 and 733. Accordingly, it is possible to reduce the pressure loss when the working gas collides with the ends 732 and 733 of each side element 73. The cross section of each side element 73 cut along the first direction is not limited to an elliptical shape, and may be any shape such as a rectangular shape. It is preferable that the ends 732 and 733 are provided in a curved shape or are inclined.

(99) FIGS. 14 and 15 are a top view and a side view of the regenerator 69 in a state where the heat storage 71 is accommodated in the housing 70. In FIG. 15, only a portion in the first direction is illustrated. As illustrated in FIGS. 14 and 15, the heat storage 71 is uniformly disposed in the circumferential direction and the axial direction (first direction) in the housing 70, and as described above, since the housing 70 and the heat storage 71 are continuously connected to each other and provided in one piece, the strength of the regenerator 69 is high.

(100) As illustrated in FIG. 14, as viewed from the first direction, the heat storage 71 includes pass-through portions 74 each of which penetrates from an end on the heater 67 side to an end on the cooler 68 side and that are regularly arranged. Specifically, each of the unit structures 72 has four through parts defined and formed in a lattice shape by the plurality of side elements 73 as viewed from the first direction, and the heat storage 71 having the pass-through portions 74 arranged in a lattice shape between the inner peripheral wall 61 and the outer peripheral wall 62 is formed by continuously providing a plurality of unit structures 72 in the first direction.

(101) In this way, since the heat storage 71 has the pass-through portions 74 penetrating in the first direction, the working gas can smoothly pass through an internal space of the heat storage 71. For example, when the heat storage is formed by stacking a plurality of metal meshes or the like as in the case of a heat storage in the related art, if the positions of the respective metal meshes are shifted, there is a possibility that the pass-through portions are not formed and clogging occurs inside the heat storage. In contrast, the pass-through portions 74 can be reliably provided in the heat storage 71 of the present embodiment.

(102) In the heat storage 71, the heat exchange amount with the working gas is larger at a position closer to the heater 67 in the first direction where the working gas has a high temperature and a position closer to the cooler 68 in the first direction where the working gas has a low temperature, and the heat exchange amount with the working gas is relatively small at a central position in the first direction.

(103) Therefore, the side elements 73 of the heat storage 71 may have different shapes depending on the positions in the first direction. For example, the side elements 73 disposed at the central portion in the first direction may have a smaller projected area S2 in the second direction than the side elements 73 disposed at both side portions of the heat storage 71 in the first direction.

(104) In the heat exchanger 6 described above, the heater 67, the cooler 68, and the regenerator 69 are continuously connected to each other and provided in one piece. Specifically, the heater 67, the cooler 68, and the regenerator 69 are continuously connected to each other and provided in one piece in additive manufacturing using a metal powder material described later.

(105) Since the heater, the cooler and the regenerator are provided in one piece, it is not necessary to separately provide the heater, the cooler, and the regenerator, to couple the heater, the cooler, and the regenerator by, for example, brazing or welding, or to provide the heater, the cooler, and the regenerator with flange portions to couple the heater, the cooler, and the regenerator by bolt

fastening at the flange portions. Accordingly, it is possible to reduce the manufacturing time and the manufacturing cost of the heat exchanger **6**. In addition, since a member for coupling is unnecessary, the size and weight of the heat exchanger **6** can be reduced.

(106) In a heat exchanger implemented by coupling the heater, the regenerator, and the cooler that are separately provided, leakage of the working gas may occur at the coupling portion. In contrast, in the present embodiment, since the heater, the regenerator, and the cooler are provided in one piece, leakage of working gas can be avoided.

(107) [Method for Manufacturing Heat Exchanger]

(108) The heat exchanger **6** according to the present embodiment is formed by additive manufacturing technology (hereinafter also referred to as AM technology) that can manufacture a component having a complicated three-dimensional shape by depositing and solidifying metal powder materials layer by layer. According to the AM technology, it is possible to manufacture a component having a fine and complicated three-dimensional shape, which is difficult to manufacture by a manufacturing method in the related art such as machining or casting.

(109) The heater **67** and the cooler **68** each have a flow channel (that is, the working gas flow channels **671** and **681**, the heat source fluid flow channel **672**, and the cooling fluid flow channel **682**) having a fine and complicated three-dimensional shape, and can be manufactured by the AM technology. In particular, the structure in which the spiral flow generation portion **55** is provided in the fine heat source fluid flow channel **672** is difficult to manufacture by a manufacturing method in the related art that does not use the AM technology.

(110) A regenerator in the related art is not manufactured using the AM technology and is formed by stacking metal meshes or the like inside a housing. The regenerator **69** of the present embodiment can be manufactured by forming the housing **70** and the heat storage **71** in one piece using the AM technology. In particular, the heat storage **71** has a fine and complicated three-dimensional shape implemented by the AM technology, and is difficult to manufacture by a manufacturing method in the related art that does not use the AM technology.

(111) As described, the heater **67**, the cooler **68**, and the regenerator **69** can be manufactured by the AM technology, and are formed in one piece by the AM technology in the present embodiment. Accordingly, as described above, a process (brazing or the like) for coupling the heater, the cooler and the regenerator is unnecessary, and the manufacturing time and the manufacturing cost of the heat exchanger **6** can be reduced.

(112) The first cylinder **23** and the ceiling wall **27** thereof provided on the inner peripheral side of the heat exchanger **6**, the heat source fluid introduction chamber **80** provided above the heater **67** and the first cylinder **23**, and the heat source fluid discharge channel **51** provided on the outer peripheral side of the heat exchanger **6** are formed in one piece with the heat exchanger **6** by the AM technology. Specifically, as illustrated in FIG. **16**, for example, a metal powder material is deposited and solidified layer by layer upward with the flange portion **25** of the first cylinder **23** as a bottom surface to perform additive manufacturing. In particular, the plurality of grooves **29** provided in the ceiling wall **27** of the first cylinder **23** have fine shapes, and are difficult to form by a manufacturing method in the related art that does not use the AM technology.

(113) Accordingly, it is unnecessary to assemble the heat exchanger **6**, the first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51**, and the manufacturing time and the manufacturing cost of the Stirling engine **1** can be reduced. The first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51** may not necessarily be formed in one piece with the heat exchanger **6**. One or two of the first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51** may be formed in one piece with the heat exchanger **6**.

(114) Further, since the fine flow channels and the heat storage **71** can be manufactured by the AM technology, the size of the heat exchanger **6** can be reduced. In the Stirling engine, when the heat exchanger is large, the dead volume of the space through which the working gas flows is large, and

the compression ratio of the working gas decreases. Therefore, the heat efficiency and the output decrease. In the present embodiment, since the size of the heat exchanger **6** is reduced, the ratio of the dead volume to the volume of the heat exchanger **6** can be reduced, and the heat efficiency and the output of the Stirling engine **1** can be improved.

(115) In order to deposit the metal powder material at an inclination angle of less than a predetermined angle (for example, less than 45 degrees) with respect to a plane intersecting with a deposition direction when forming a component having a predetermined size by additive manufacturing, it is necessary to simultaneously form a support member that supports the inclined portion from below by additive manufacturing. When the support member is formed, it is necessary to remove the support member, for example, manually after component manufacturing.

(116) In the present embodiment, in the heat exchanger **6**, the first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51**, an inclination angle equal to or greater than the predetermined angle (for example, 45 degrees) is appropriately set for a portion deposited on the upper side in an inclined manner, and formation of the support member is not necessary at the time of additive manufacturing. As illustrated in FIG. **16**, for example, in the wall surface of the heat source fluid discharge channel **51** and the discharge portion **52** of the heat source fluid discharge channel **51**, the inclination angle is set to 45 degrees for portions deposited on the upper side in an inclined manner. Accordingly, the formation of the support member can be avoided.

(117) [Modification]

(118) Although the Stirling engine **1** has been described as an example of the Stirling machine of the present invention in the embodiment described above, the present invention is not limited thereto. For example, the Stirling machine of the present invention may be a heat pump using a reverse Stirling cycle. Specifically, the heat pump serving as a Stirling machine may be configured such that the electric generator **11** of the Stirling engine **1** described above functions as a motor that inputs mechanical power to the piston crank mechanism **5** to cause the displacer piston **3** and the power piston **4** to reciprocate.

(119) When the heat exchanger **6** is applied to the heat pump serving as a Stirling machine, the electric generator **11** functions as a motor, so that a fluid flowing through the heater **67** receives heat from the working gas and becomes a high-temperature fluid, and a fluid flowing through the cooler **68** has its heat absorbed by the working gas and becomes a low-temperature fluid. In the present embodiment, with a fine internal structure, the heat exchanger **6** has a reduced dead volume and high heat exchange performance, and thus the fluid can be heated to an ultra-high temperature or cooled to an extremely low temperature.

(120) Although an embodiment of the present invention has been described above with reference to the accompanying drawings, it is needless to say that the present invention is not limited to the embodiment. It is apparent to those skilled in the art that various changes or modifications can be conceived within the scope described in the claims, and it is understood that the changes or modifications naturally fall within the technical scope of the present invention. In addition, the components in the above embodiment may be freely combined without departing from the gist of the invention.

(121) For example, although the β -type Stirling engine **1** has been described as an example of a Stirling machine to which the heat exchanger **6** is applied in the embodiment described above, the present invention is not limited thereto. The heat exchanger **6** may be applied to a Stirling machine having another form, specifically, an α -type Stirling engine, a γ -type Stirling engine, or the like.

(122) Further, although an example has been described in which the regenerator **69** is applied to the Stirling engine **1** that is an example of a Stirling machine in the embodiment described above, the present invention is not limited thereto. The regenerator **69** can be applied to various heat engines.

(123) Although the side element **73** constituting the heat storage **71** has a plate shape in the embodiment described above, the present invention is not limited thereto. The side element **73** may

be a linear body.

(124) In the present description, at least the following matters are described. In the parentheses, corresponding components and the like in the above embodiment are illustrated as an example, but the present invention is not limited thereto.

(125) (1) A heat exchanger (heat exchanger **6**) for a Stirling machine (Stirling engine **1**), the Stirling machine including an expansion chamber (expansion chamber **21**) and a compression chamber (compression chamber **22**), the heat exchanger including: a heater (heater **67**); a regenerator (regenerator **69**); and a cooler (cooler **68**), in which the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid (working gas), and communicate the expansion chamber with the compression chamber, the heater includes a first flow channel (working gas flow channel **671**) communicating with the expansion chamber and the regenerator and configured to allow the working fluid to flow therethrough, and a second flow channel (heat source fluid flow channel **672**) configured to allow a first fluid (heat source fluid) for exchanging heat with the working fluid to flow therethrough, the cooler includes a third flow channel (working gas flow channel **681**) communicating with the compression chamber and the regenerator and configured to allow the working fluid to flow therethrough, and a fourth flow channel (cooling fluid flow channel **682**) configured to allow a second fluid (cooling fluid) for exchanging heat with the working fluid to flow therethrough, and the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece.

(126) According to (1), since the heater, the regenerator, and the cooler are provided in one piece by being continuously connected to each other, it is not necessary to perform brazing, bolt fastening at flange portions, or the like in order to couple the heater, the regenerator, and the cooler.

Accordingly, it is possible to reduce the manufacturing time and the manufacturing cost of the heat exchanger. Further, occurrence of leakage of the working gas can be avoided.

(127) (2) The heat exchanger for a Stirling machine according to (1), in which the regenerator includes a heat storage (heat storage **71**) that is capable of storing heat of the working fluid, and a housing (housing **70**) that surrounds the heat storage, and the heat storage and the housing are continuously connected to each other, provided in one piece.

(128) According to (2), since the heat storage and the housing of the regenerator are provided in one piece by being continuously connected to each other, a complicated structure having high heat exchange efficiency can be adopted as the heat storage. Further, as compared with a regenerator in which a plurality of metal meshes or the like are integrated into a housing, the number of assembly steps of the regenerator can be reduced.

(129) (3) The heat exchanger for a Stirling machine according to (2), in which the heat storage includes a plurality of linear bodies or plate-shaped bodies (side elements **73**) that are formed in a three-dimensional manner and in one piece.

(130) According to (3), since the heat storage includes a plurality of linear bodies or plate-shaped bodies that are formed in a three-dimensional manner and in one piece, it is possible to reduce a dead volume that is a volume not involved in heat exchange in the housing, and it is possible to improve the heat storage performance of the regenerator.

(131) (4) The heat exchanger for a Stirling machine according to any one of (1) to (3), in which the heater, the regenerator, and the cooler are an additive manufactured body formed by depositing a metal powder.

(132) According to (4), the heater, the regenerator, and the cooler can be formed in one piece by additive manufacturing.

(133) (5) A method for manufacturing the heat exchanger according to any one of (1) to (3), including: forming the heater, the cooler, and the regenerator in one piece by performing additive manufacturing using a metal powder.

(134) According to (5), the heater, the regenerator, and the cooler can be formed in one piece by additive manufacturing.

(135) (6) A Stirling machine (Stirling engine **1**) including: a heat exchanger (heat exchanger **6**) that includes a heater (heater **67**), a regenerator (regenerator **69**), and a cooler (cooler **68**), which are connected in series in a flow direction of a working fluid (working gas); and a cylinder (first cylinder **23**) that includes an expansion chamber (expansion chamber **21**) connected to the heater and a compression chamber (compression chamber **22**) connected to the cooler, in which the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger, the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece, and the heat exchanger and the cylinder are continuously connected to each other, provided in one piece.

(136) According to (6), since the heater, the regenerator, and the cooler are provided in one piece by being continuously connected to each other, it is not necessary to perform brazing, bolt fastening at flange portions, or the like in order to couple the heater, the regenerator, and the cooler. In addition, since the heat exchanger and the cylinder are provided in one piece by being continuously connected to each other, a process of coupling the heat exchanger and the cylinder is also unnecessary. Thus, it is possible to reduce the manufacturing time and the manufacturing cost of the Stirling machine.

(137) (7) The Stirling machine according to (6), in which the heat exchanger and the cylinder are an additive manufactured body formed by depositing a metal powder.

(138) According to (7), the heat exchanger and the cylinder can be formed in one piece by additive manufacturing.

(139) (8) The Stirling machine according to (6) or (7), further including: a displacer piston (displacer piston **3**) and a power piston (power piston **4**) configured to reciprocate; and a crankshaft (crankshaft **10**) coupled to the displacer piston and the power piston, in which the expansion chamber is a space defined by the displacer piston and the cylinder, the compression chamber is a space defined by the displacer piston, the power piston, and the cylinder, and the crankshaft is connected to an electric motor or an electric generator (electric generator **11**).

(140) According to (8), a Stirling machine including a heat exchanger and a cylinder that are formed in one piece can be implemented.

Claims

1. A heat exchanger for a Stirling machine, the Stirling machine including an expansion chamber and a compression chamber, the heat exchanger comprising: a heater; a regenerator; and a cooler, wherein the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid, and communicate the expansion chamber with the compression chamber, the heater includes a first flow channel communicating with the expansion chamber and the regenerator and configured to flow the working fluid therethrough, and a second flow channel configured to flow a first fluid for exchanging heat with the working fluid to flow therethrough, the cooler includes a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and the heater, the regenerator, and the cooler are disposed between a radially inner peripheral wall and a radially outer peripheral wall and the heater, the regenerator, and the cooler are continuously connected to each other in an axial direction of the radially inner peripheral wall and the radially outer peripheral wall and configured as one additively manufactured piece such that a separate coupling of the heater, the regenerator, and the cooler is not necessary to couple each of the heater, the regenerator, and the cooler to each other.

2. The heat exchanger for the Stirling machine according to claim 1, wherein the regenerator

- includes a heat storage that is capable of storing heat of the working fluid, and a housing that surrounds the heat storage, and the heat storage and the housing are continuously connected to each other, provided in one second piece.
3. The heat exchanger for the Stirling machine according to claim 2, wherein the heat storage includes a plurality of linear bodies or a plurality of plate-shaped bodies, the plurality of linear bodies and the plurality of plate-shaped bodies are each formed in a three-dimensional manner and in one third piece.
4. The heat exchanger for the Stirling machine according to claim 3, wherein the expansion chamber and the compression chamber are formed inside a cylinder of the Stirling machine, and arranged in series along an axial direction of the cylinder interposing a piston therebetween, and the radially inner peripheral wall and the radially outer peripheral wall extend in the axial direction of the cylinder along an outer peripheral surface of the cylinder.
5. The heat exchanger for the Stirling machine according to claim 1, wherein the heater, the regenerator, and the cooler are an additive manufactured body formed by depositing a metal powder.
6. A method for manufacturing the heat exchanger according to claim 1, comprising: forming the heater, the cooler, and the regenerator in the one piece by performing additive manufacturing using a metal powder.
7. A Stirling machine comprising: the heat exchanger of claim 1; and a cylinder including an expansion chamber connected to the heater and a compression chamber connected to the cooler, wherein the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger.
8. The Stirling machine according to claim 7, wherein the heat exchanger and the cylinder are an additive manufactured body formed by depositing a metal powder.
9. The Stirling machine according to claim 7, further comprising: a displacer piston and a power piston configured to reciprocate; and a crankshaft coupled to the displacer piston and the power piston, wherein the expansion chamber is a space defined by the displacer piston and the cylinder, the compression chamber is a space defined by the displacer piston, the power piston, and the cylinder, and the crankshaft is connected to an electric motor or an electric generator.
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